

Good question — this is one of the **core concepts** in networking, and once you understand it, routing tables become much easier.

Let's go step by step, slowly and clearly.

◆ What is the “Destination” in a Routing Table?

In a routing table, each entry looks like:

Destination	Gateway	Interface
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👉 The **destination** is **NOT** a single IP (most of the time)

👉 It is a **network address + mask (CIDR)**

◆ So how do we calculate

the destination?

You always compute it like this:

`Destination = Network Address`

And the network address comes from:

`IP address + Subnet Mask`

◆ Step-by-step method (very important)

Let's say you have:

IP: 192.168.1.34
Mask: 255.255.255.0 (/24)

Step 1: Convert mask mentally

255.255.255.0 = /24

👉 Means first 24 bits = network

👉 Last 8 bits = host

Step 2: Find the network part

Keep network bits, zero the rest:

```
192.168.1.34  
→ 192.168.1.0
```

 **Final result:**

```
Destination = 192.168.1.0/24
```

◆ Why not use the full IP?

Because routing works like this:

- 👉 Routers don't route to ONE device
- 👉 They route to a **network**

So instead of:

```
192.168.1.34 ❌
```

We use:

192.168.1.0/24 

◆ Another example (important)

IP: 10.0.5.23

Mask: 255.0.0.0 (/8)

 Network part = first 8 bits

10.0.5.23 → 10.0.0.0

 **Destination:**

10.0.0.0/8

◆ Special case: Default Route

Sometimes you see:

0.0.0.0/0

👉 This means:

- “If nothing matches, send here”

◆ Key idea (very important for NetPractice)

When a device sends a packet:

1. It checks all destinations in the routing table
2. It picks the **most specific match** (longest mask)

Example:

```
192.168.1.0/24    ← more specific
192.168.0.0/16    ← less specific
```

👉 If IP = 192.168.1.10, it picks /24

◆ Now I guide you (don't jump to answer)

Try this yourself:

IP: 172.16.45.10
Mask: 255.255.0.0

👉 Ask yourself:

1. What is the CIDR?
2. Which part is network?
3. What happens if I zero the host bits?

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If you want, I can check your answer or walk you through it interactively 👍